Order of Operations with Radicals

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Evaluate multi-step expressions with rational numbers, exponents and radicals, in six steps or fewer.

Must not No mnemonic. The state guide says to avoid PEMDAS because it leaves out grouping symbols, and the word appears on no student page in this unit.

Target I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

MTR **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 3 and 7 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. There is nothing to memorise. Read the expression, ask what is holding what, and the order tells you itself.
2. Nothing says multiply before divide. They are one level, taken left to right like a sentence.
3. A fraction bar is a grouping symbol on the top and another one on the bottom.

Slide by Slide

1. **Order of Operations with Radicals.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite and it is the one half the room gets wrong. *1 retrieves 4.05 — finish under the bar first: 25 − 9 = 16 (yesterday). 2 retrieves 4.04 — the middle zero counts and the trailing zero counts, because a decimal point is written (last week). 3 retrieves 3.05 — a negative exponent flips the base (last unit). 4 retrieves MA.5.AR.2.2 — multiplication and division are one level, taken left to right (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 4 2. 4 3. 9/4 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** There is nothing to memorise here. Read the expression, ask what is holding what, and the order tells you itself.
6. **Notes.** Nothing in mathematics says multiply before divide. They are the same level, and you read left to right like a sentence.
7. **Notes.** A fraction bar is a grouping symbol on the top and another one on the bottom. Finish each, then divide.
8. **Example 1 — A grouping symbol inside a grouping symbol.** SPOKEN ONLY.
9. **Example 1 — Worked.** The power is inside the parentheses, so it goes first even though it is a level below grouping — because it is inside.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, a square root and a positive answer.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A fraction bar and a radical in one expression.** SPOKEN ONLY.
13. **Example 2 — Worked.** Three groups, each finished on its own before anything crosses between them.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Same three groups, and both halves land on a half.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **16–17. Board 1.** $90 \div 5 \cdot 3$ → ***54*** *6 — 5 · 3 was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]; 1,350 — the division was read as a multiplication*
17. **18–19. Board 2.** A square tile has side √196 centimeters. Its diagonal is about 19.8 centimeters. What is the perimeter of the tile? → ***56 centimeters*** *The diagonal of about 19.8 centimeters is not needed. anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]; 14 — the side, not the perimeter; 196 — the root was never taken*
18. **20–21. Board 3.** $16 - 5 + 4$ → ***15*** *7 — 5 + 4 was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]*
19. **22–23. Board 4.** $\frac{3^3 - 21}{5^2 - 15}$ → ***3/5*** *anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]; −6/5 — 5^2 read as 5 · 2, leaving 6 over −5; 6/25 — the −15 under the bar was dropped*

24–25. Board 5. $2 + 3 \cdot 4^2$. Which is the value? → ***B — 50*** *(A) the exponent was applied to 3 · 4 instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]; (C) 4^2 was read as 4 · 2 [8.NSO.1.3: an exponent is not a factor]; (D) 2 + 3 was done first, then 5 · 16 [8.NSO.1.7: multiplication comes before addition] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. $(6^2 - 4)$ over 2^3. Which is the value? → ***A — 4*** *(B) only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]; (C) 6^2 was divided by 2^3 first and then 4 was subtracted [8.NSO.1.7: the bar groups the whole numerator]; (D) the numerator was multiplied by 2^3 instead of divided [8.NSO.1.7: the bar is division] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $-2^2 + (-2)^2$ → ***0*** *8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]; −8 — both terms read as -2^2*

30–31. Board 8. $12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$ → ***17/6*** *12/8 + 12/9 left unadded — a value, not an answer; 1/8 + 1/9 = 2/17 — the denominators were added [Grade 5: adding fractions needs a common denominator]*

32–33. Board 9. Roscoe says that $(\sqrt{n} - 11)^2$ is always positive, because a square is always positive. Is he right? Give a value of n that settles it. → ***No — at n = 121 the value is 0*** *'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]; 'no' with no value of n — the counterexample is the evidence*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–6 are one structural decision each: multiply before adding, left-to-right division, a power inside parentheses, a fraction bar, the minus-sign-and-exponent contrast, and a radical with a negative exponent. 7 is the three items the state assessment supplies and 8 is three of our own with a bracket, a fraction bar and a cube root. 9 works backwards in three shapes. The pattern breaks at 10, where the student judges a rule rather than a value. 11 is the diagnostic multiple choice, whose three wrong options are three different levels misapplied, and 12 is the select-all. The Additional sheet mirrors each position, and its question 10 is the same error one level down — addition and subtraction rather than multiplication and division.

Misconceptions to Watch

Board 1. 6 — 5 · 3 was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]

Board 2. anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]

Board 3. 7 — 5 + 4 was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]

Board 4. anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]

Board 5. the exponent was applied to 3 · 4 instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]

Board 6. only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]

Board 7. 8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]

Board 8. 12/8 + 12/9 left unadded — a value, not an answer

Board 9. 'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]

Materials Whiteboards and markers. Calculators are allowed on everything, and this is the lesson where a calculator is worth using against the class: enter -2^2 and $(-2)^2$ and let the machine settle it.